Exact vs ours on graph classes with exact algorithms

Trees
Plot 1 · |S|, unit weights
Exact (tree DP)
Ours (GNN+RL + RR)
Greedy + RR
Ours optimal in 92% of graphs
|S|
Number of nodes (n)

Unit-interval graphs
Plot 1 · |S|, unit weights
Ours optimal in 32% of graphs
|S|
Number of nodes (n)

Convex bipartite graphs
Plot 1 · |S|, unit weights
Ours optimal in 52% of graphs
|S|
Number of nodes (n)

2-trees
Plot 1 · |S|, unit weights
Ours optimal in 89% of graphs
|S|
Number of nodes (n)

Plot 2 · |S|, random weights
Exact (tree DP)
Ours (GNN+RL + RR)
Greedy + RR
Ours optimal (W) in 49% of graphs
same |S| as exact in 78%
|S|
Number of nodes (n)

Plot 2 · |S|, random weights
Ours optimal (W) in 22% of graphs
same |S| as exact in 33%
|S|
Number of nodes (n)

Plot 2 · |S|, random weights
Ours optimal (W) in 28% of graphs
same |S| as exact in 42%
|S|
Number of nodes (n)

Plot 2 · |S|, random weights
Ours optimal (W) in 30% of graphs
same |S| as exact in 41%
|S|
Number of nodes (n)

Plot 3 · W(S), random weights
Exact (tree DP)
Ours (GNN+RL + RR)
Greedy + RR
Ours optimal in 49% of graphs
Total weight W(S)
Number of nodes (n)

Plot 3 · W(S), random weights
Ours optimal in 22% of graphs
Total weight W(S)
Number of nodes (n)

Plot 3 · W(S), random weights
Ours optimal in 28% of graphs
Total weight W(S)
Number of nodes (n)

Plot 3 · W(S), random weights
Ours optimal in 30% of graphs
Total weight W(S)
Number of nodes (n)

Row 2: With weighted graphs, several optimal sets can have the same minimum weight but different sizes, so |S| can differ from the exact method even when ours is optimal. See Plot 3.